Fluid mechanics – liquids and gases in motion

The air around an aircraft wing, the water around a ship, the blood in our veins, the weather and the ocean currents – all follow the same equations. Fluid mechanics is one of the oldest branches of physics and at the same time one of the most unfinished: the Navier–Stokes equations have been known since 1845, but whether they always have well-behaved solutions is one of the seven Millennium Prize Problems of mathematics, and turbulence is still called the last great unsolved problem of classical physics. This page covers the foundations with three interactive figures and points to the site's simulations of wind tunnels and towing tanks.

Hydrostatics

Around 250 BC Archimedes found that a body in a fluid experiences an upthrust equal to the weight of the fluid it displaces. In 1648 Blaise Pascal showed that the pressure in a fluid at rest is transmitted equally in all directions, and that it increases with depth:

$$p = p_0 + \rho g h, \qquad F_{\text{buoyancy}} = \rho_{\text{fluid}}\, V g$$

At a depth of 10 metres of water the pressure has doubled. Pascal's principle is the basis of hydraulic presses and brakes, and Torricelli's barometer measures the pressure of the atmosphere as a column of liquid (see Torricelli).

Continuity and Bernoulli's equation

Mass can neither appear nor disappear. For flow through a pipe at constant density this means that the volume flow rate is the same everywhere: $A_1 v_1 = A_2 v_2$ – the water runs faster where the pipe is narrow. In general this is expressed by the continuity equation

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{v}) = 0 \qquad \Rightarrow \qquad \nabla \cdot \vec{v} = 0 \ (\text{incompressible})$$

In 1738 Daniel Bernoulli found that conservation of energy along a streamline in an ideal fluid gives

$$p + \tfrac{1}{2}\rho v^2 + \rho g z = \text{const.}$$

Where the speed is high, the pressure is low. This explains the Venturi tube, the atomiser and the Pitot tube that measures the speed of aircraft. For the lift on an aircraft wing, though, Bernoulli's equation is only half the story – see below.

Euler and the ideal fluid

In 1757 Leonhard Euler wrote down the equation of motion for a fluid without friction – Newton's second law for a small element of fluid:

$$\rho\left(\frac{\partial \vec{v}}{\partial t} + (\vec{v}\cdot\nabla)\vec{v}\right) = -\nabla p + \rho\vec{g}$$

The term $(\vec{v}\cdot\nabla)\vec{v}$ is non-linear: the velocity transports itself. It is the source of almost all the difficulties of fluid mechanics – and of its richness. For flows without vortices the velocity can be written as the gradient of a potential, $\vec{v} = \nabla\phi$, and then $\nabla^2\phi = 0$ – Laplace's equation. But the ideal theory has an embarrassing problem: in 1752 d'Alembert showed that a body moving at constant speed through an ideal fluid feels no resistance at all. That contradicted all experience.

The Navier–Stokes equations

In 1822 Claude-Louis Navier added a term for internal friction, viscosity, and in 1845 George Gabriel Stokes gave a clear derivation. For an incompressible Newtonian fluid with kinematic viscosity $\nu = \eta/\rho$:

$$\frac{\partial \vec{v}}{\partial t} + (\vec{v}\cdot\nabla)\vec{v} = -\frac{1}{\rho}\nabla p + \nu\nabla^2\vec{v} + \vec{g}, \qquad \nabla\cdot\vec{v} = 0$$

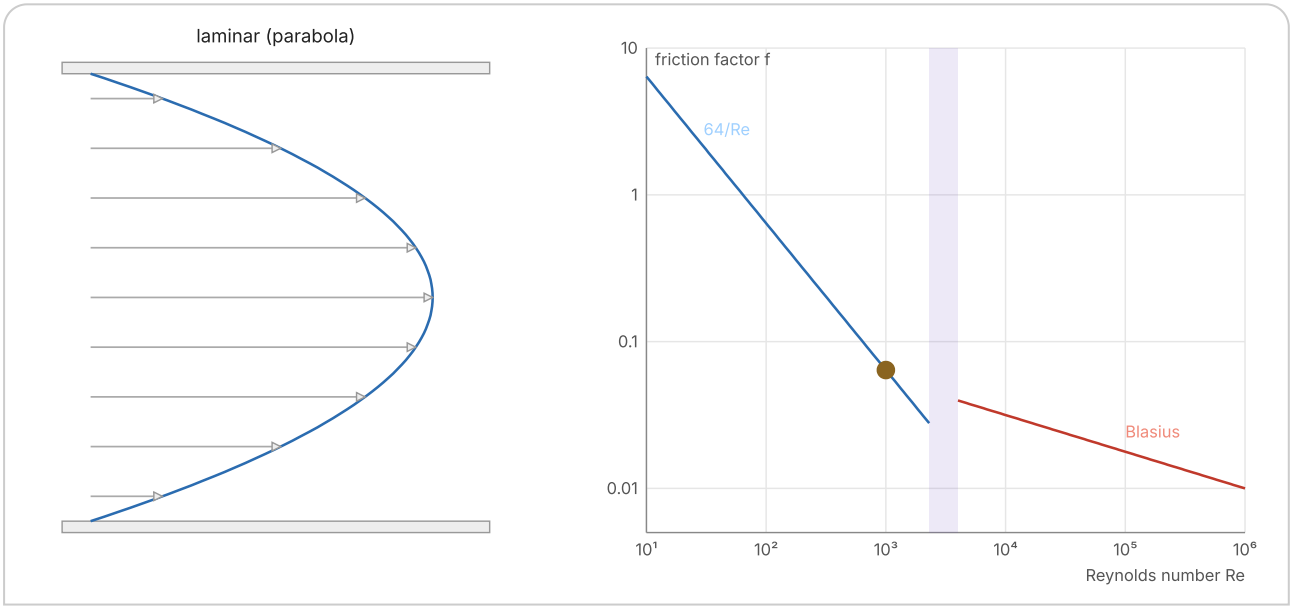
The equations describe everything from blood flow to jet streams. But no one knows whether smooth initial conditions in three dimensions always give smooth solutions for all time – since 2000 the Clay Mathematics Institute has offered a million dollars for a proof or a counterexample. In practice the equations are solved numerically, as in the Navier–Stokes wind tunnel on this site.

The Reynolds number

In 1883 in Manchester Osborne Reynolds let water flow through a glass pipe and injected a thin streak of dye into the middle. At low speed the streak stayed a sharp line – laminar flow. At higher speed it suddenly began to waver and was whirled out across the whole pipe – turbulence. The transition depended on just one dimensionless quantity:

$$\mathrm{Re} = \frac{\rho v D}{\eta} = \frac{v D}{\nu}$$

The Reynolds number is the ratio of inertial to viscous forces. In pipes the flow is laminar below Re ≈ 2300. Two flows with the same Re and the same geometry are alike – which is why ships can be tested with models in a towing tank and aircraft in a wind tunnel. A swimming bacterium has Re ≈ 10^{-5} and lives in a world without inertia; a blue whale has Re ≈ 10^{8}.



The boundary layer

The resolution of d'Alembert's paradox came from Ludwig Prandtl in an eight-page lecture at Heidelberg in 1904. Even when the viscosity is small, it cannot be ignored completely: at a solid surface the fluid has zero velocity ("no slip"). Friction acts only in a thin *boundary layer* near the surface, of thickness $\delta \sim L/\sqrt{\mathrm{Re}}$; outside it the flow is almost ideal. Prandtl's student Heinrich Blasius solved the boundary-layer equations for a flat plate in 1908. Behind a body the boundary layer can separate, and then a wake with vortices and pressure drag forms. The dimples on a golf ball make the boundary layer turbulent, so it stays attached longer and the wake becomes smaller – see The flight of the ball.

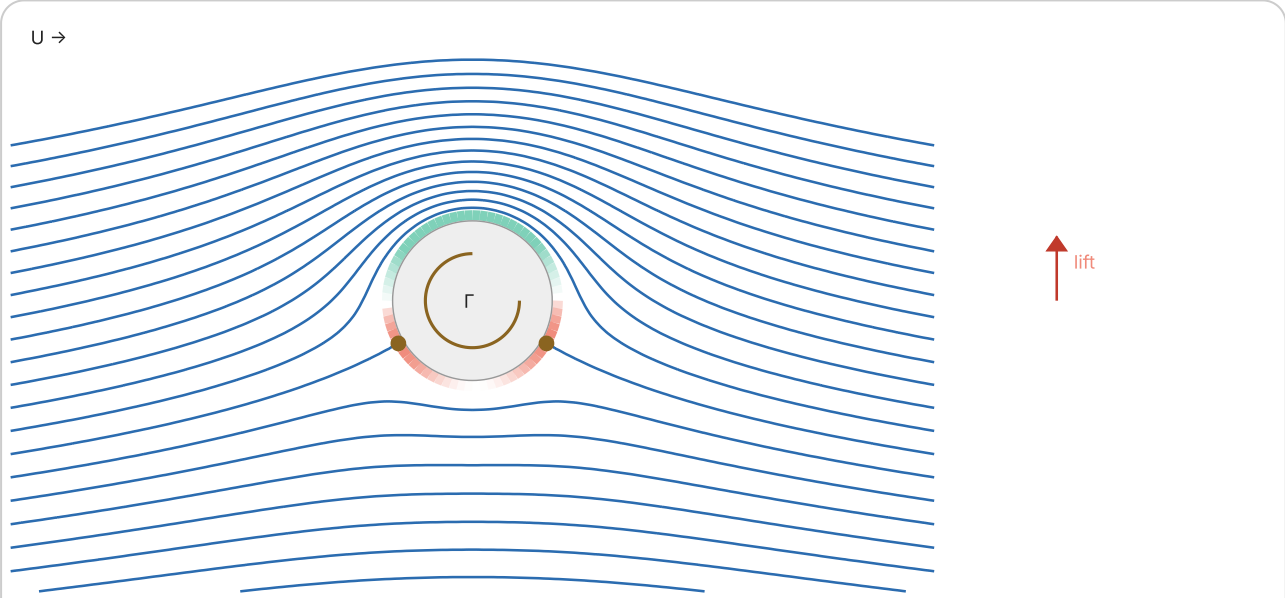
$$F_D = \tfrac{1}{2} C_D \, \rho v^2 A$$

Lift and circulation

Why does a wing fly? Wilhelm Kutta (1902) and Nikolai Zhukovsky (Joukowski, 1906) showed that the lift per unit length of the wing is proportional to the *circulation* Γ – how much the air as a whole turns round the wing:

$$L' = \rho U \Gamma, \qquad \Gamma = \oint \vec{v} \cdot d\vec{l}$$

The sharp trailing edge of the wing forces the flow to leave it smoothly (the Kutta condition), and that fixes the circulation. The air over the wing moves faster (lower pressure, Bernoulli), and the wing deflects the air downwards (Newton's third law) – the two explanations are two sides of the same thing. The same formula explains the Magnus effect: a spinning ball or cylinder in a flow experiences a sideways force. In 1924 Anton Flettner built a ship, the Buckau, which sailed with rotating cylinders instead of sails.



Vortices

In 1858 Hermann von Helmholtz showed that vortex lines in an ideal fluid move with the fluid and can neither be created nor destroyed. On this Kelvin built a (wrong, but beautiful) theory of atoms as vortex knots in the ether. Behind a cylinder in a real fluid the vortices are shed alternately from the upper and lower sides and form a Kármán vortex street (Theodore von Kármán, 1911). The frequency is given by the Strouhal number, $St = fD/U \approx 0.2$ – which is

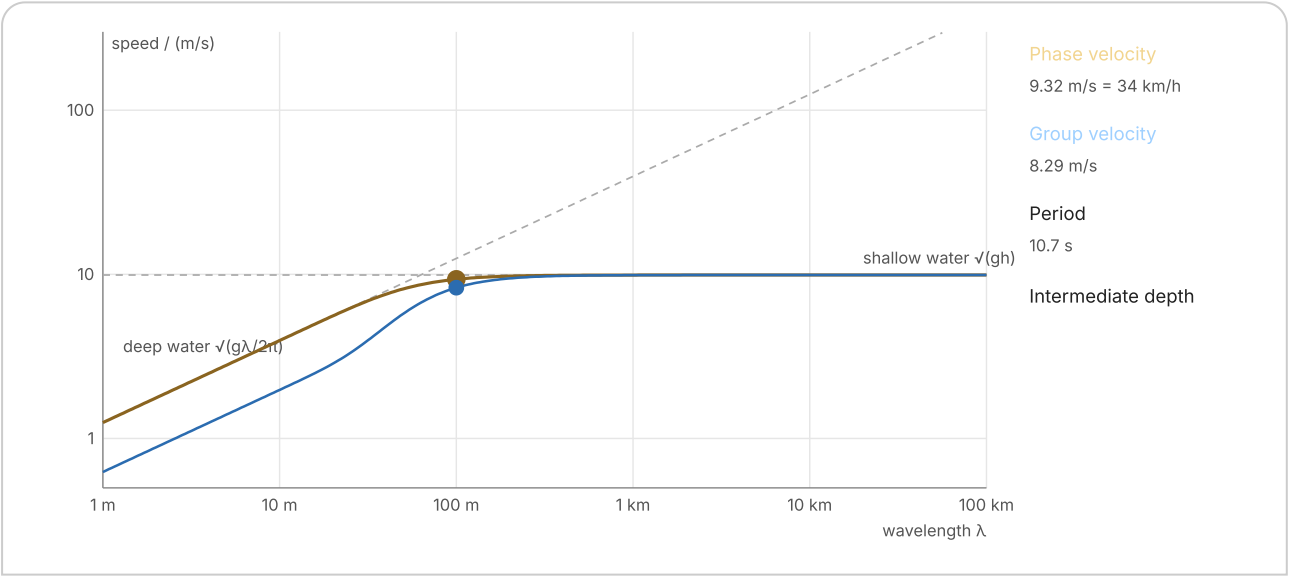
what makes telephone wires sing in the wind, and what set the Tacoma Narrows Bridge oscillating before it collapsed in 1940. See the wind tunnel and The V formation.

Water waves

Waves on water are an interplay between gravity and the inertia of the water. For waves with wave number $k = 2\pi/\lambda$ on water of depth h the dispersion relation is

$$\omega^2 = gk\tanh(kh), \qquad c = \frac{\omega}{k}, \qquad c_g = \frac{d\omega}{dk}$$

In deep water $c = \sqrt{g\lambda/2\pi}$ – long waves run fastest, and the group velocity is half the phase velocity. That is why swell from a distant storm arrives with the longest waves first. In shallow water $c = \sqrt{gh}$, independent of wavelength: a tsunami in 4000 metres of water travels at 700 km/h (see Tsunami). A ship makes waves moving at the same speed as the ship; when the wavelength equals the length of the ship, the ship "climbs" its own bow wave – this is *hull speed*, $v \approx 1.34\sqrt{L/\text{ft}}$ knots (see The towing tank and Cross seas).



Turbulence

In 1922 Lewis Fry Richardson described turbulence in a rhyme: big whirls have little whirls that feed on their velocity, and little whirls have lesser whirls, and so on to viscosity. In 1941 Andrey Kolmogorov made the idea quantitative: energy is fed in at large scales, transferred without loss to smaller and smaller eddies, and dissipated as heat only at the smallest scale, $\eta_K = (\nu^3/\varepsilon)^{1/4}$. In the intermediate range the energy is distributed over wave numbers as

$$E(k) = C\,\varepsilon^{2/3} k^{-5/3}$$

Kolmogorov's −5/3 law has been measured in tidal currents, wind tunnels and the atmosphere. But the full statistics of turbulence, with its intermittent bursts, is still not understood. Werner Heisenberg is said to have remarked that he would ask God two questions: why relativity, and why turbulence – and that he expected an answer to the first.

Further reading

- D.J. Tritton: *Physical Fluid Dynamics* (2nd ed., Oxford 1988)
- G.K. Batchelor: *An Introduction to Fluid Dynamics* (Cambridge 1967)
- Richard Feynman: *The Feynman Lectures on Physics*, vol. II, ch. 40–41: "The Flow of Dry Water" and "The Flow of Wet Water"
- Tom Mullin: *Fluid Dynamics. A Very Short Introduction* (Oxford 2017)

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